

Affan Moyeed

Professional Summary:

- **Experienced Senior AI Full Stack Engineer** with 7+ years of expertise in designing, developing, testing, and deploying robust **web and application solutions** using **Python, Flask, React, and PostgreSQL**.
- Strong expertise in **cloud computing** with **Azure Kubernetes Service (AKS), Azure Function Apps, Azure Blob Storage, and Azure Key Vault**, ensuring scalable and secure enterprise applications.
- Proficient in **CI/CD pipeline automation** using **GitHub Actions, Terraform, and Docker**, enhancing deployment efficiency and maintaining high software quality.
- Skilled in **data engineering and ETL pipelines**, leveraging **Snowflake, Azure Synapse Analytics, and Azure Data Factory** for efficient data processing, storage, and analytics.
- Extensive experience in **API development**, including **RESTful APIs, FastAPI, and GraphQL**, with a focus on high-performance, scalable microservices architecture.
- Strong understanding of **authentication and security** protocols such as **OKTA, OAuth2, JWT, and Azure Key Vault**, ensuring compliance with enterprise security standards.
- Expertise in **database architecture, query optimization, and data modeling** using **PostgreSQL, MySQL, and Snowflake**, improving system performance and reducing latency.
- Hands-on experience in **DevOps practices** including **containerization, Kubernetes cluster management, Helm Charts, and infrastructure automation**.
- Well-versed in **automated testing** using **pytest**, developing **unit, integration, and regression tests** to maintain system stability and reliability.
- Skilled in **logging and monitoring** using **Azure Application Insights, Prometheus, and Grafana**, ensuring proactive issue resolution and system observability.
- Successfully **migrated an enterprise application from Dev to UAT to Prod**, working closely with **DBA, Security, and Business teams** to optimize application performance, security, and compliance.
- Experienced in **business intelligence and visualization tools** like **Power BI and Tableau**, creating insightful dashboards for data-driven decision-making.
- Strong communication and collaboration skills, working with **cross-functional teams, business stakeholders, and product managers** to deliver high-quality enterprise solutions.

Technical Skills:

Programming Languages	Python 3.x, C#, JavaScript, SQL
Frameworks	Flask, Django, Quartz, React, Next.js, Node.js, Angular.js, JQuery
Databases	PostgreSQL, Snowflake, MySQL, MongoDB, CosmoDB,
Data Warehousing	Snowflake, Azure Synapse Analytics
Cloud Services	Azure Kubernetes Service (AKS), Azure Function App, Azure Blob Storage, Azure Key Vault
CI/CD & DevOps	GitHub Actions, Docker, Terraform, Jenkins
Authentication & Security	OKTA, OAuth2, JWT, Azure Key Vault
Infrastructure as Code (IaC)	Terraform, Kubernetes (K8s), Helm Charts
Testing & Automation	Pytest, Unit Testing, Integration Testing
Logging & Monitoring	Azure Application Insights, Prometheus, Grafana
API Development	RESTful APIs, FastAPI, GraphQL
ETL & Data Pipelines	Azure Data Factory, Azure Function App, ETL Pipelines
Version Control	Git, GitHub, Azure DevOps
Business Intelligence & Visualization	Power BI, Tableau
Operating Systems	Linux (Ubuntu, RHEL), Windows, Mac

Professional Experience:

Client: Mckesson, Dallas, Texas

Nov 2024 – Till date

Role: Sr. AI Full Stack Software Engineer

Responsibilities:

- Developed and maintained a large-scale enterprise application using Python (Flask for backend) and React (frontend) with PostgreSQL as the database. Optimized Flask routes, SQL queries, and API response times to improve system efficiency.
- Designed and deployed the application on Azure Kubernetes Service (AKS) with a well-structured CI/CD pipeline using GitHub Actions. Implemented Docker-based microservices for seamless scaling and efficiency.
- Migrated the project across Dev, UAT, and Prod environments. Worked closely with DevOps teams to ensure a secure and scalable Kubernetes namespace setup for all environments.
- Developed ETL processes using Azure Function App, ensuring smooth data ingestion and transformation. Managed Snowflake as the data warehouse and integrated pipelines to maintain high data quality.
- Worked with the OKTA team to implement secure authentication mechanisms, including certificate-based authentication and API security. Configured Azure Key Vault for storing environment secrets securely.
- Collaborated with the DBA team to implement a secure and optimized database architecture, improving query performance and indexing strategies.
- Wrote comprehensive unit and integration tests for the entire backend using pytest, ensuring high code coverage and system reliability before UAT sign-off.
- Worked closely with business users to implement last-minute client requirements and ensured smooth UAT sign-off. Led agile development cycles, actively collaborating with product managers and stakeholders.
- Set up and managed GitHub Secrets and Azure Key Vault for environment variables. Automated deployments using Terraform, enhancing infrastructure management across different environments.
- Integrated Azure Application Insights and Prometheus/Grafana for proactive monitoring and logging, ensuring system health and quick debugging.
- Developed custom Flask-based API integrations with internal systems, automating repetitive tasks and reducing manual efforts in business workflows.

Client: Microsoft, Redmond, WA

Sep 2023 – Oct 2024

Role: Sr. Software Engineer

- Spearheaded the development of a sophisticated Quartz application designed for robust backend processing and API integration, significantly improving process efficiency and system responsiveness.
- Led the development and integration of complex logic for an internal Microsoft team chatbot using Python. Enhanced chatbot functionalities with OpenAI technologies, enabling advanced conversation handling and dynamic response generation.
- Implemented cutting-edge search functionality and document intelligence capabilities using Azure Cognitive Search, optimizing search queries and implementing AI to effectively retrieve and understand document content.
- Utilized libraries such as requests and http.client to build and manage RESTful APIs, ensuring secure and efficient data exchange over HTTP/HTTPS protocols. Implemented OAuth2 and JWT for secure authentication and authorization.
- Integrated API gateways (e.g., Kong, NGINX) and load balancers to manage traffic, ensuring efficient routing, rate limiting, and security. Designed and implemented custom middleware to handle cross-cutting concerns.
- Expert in utilizing Cosmos DB for robust database management and connectivity, ensuring optimal performance, scalability, and data consistency across distributed systems.
- Managed and encrypted keys and sensitive data using Azure Key Vault, adhering to stringent security protocols and ensuring compliance with data protection regulations.
- Developed and deployed Azure Functions, leveraging Service Bus Queue to orchestrate and automate workflows, reducing manual intervention and streamlining operations.

- Engineered and maintained high-performance APIs to support scalable applications, ensuring robust integration with other services and platforms for seamless data exchange and system interoperability.
- Proficient in Next.js, leveraging its server-side rendering capabilities to build fast and SEO-friendly applications. Skilled in creating static and dynamic sites with Next.js, utilizing its robust routing system and API routes for serverless functions.
- Strong background in optimizing Nuxt.js applications for performance and scalability, employing techniques like static site generation (SSG) and server-side rendering (SSR). Experienced in integrating Nuxt.js with headless CMSs, providing seamless content management solutions.
- Utilized Azure Synapse Analytics for comprehensive data analysis, writing Python scripts for data manipulation and transformation to derive actionable insights and enhance decision-making processes.
- Conducting research and development on integrating tabular data and matrices with OpenAI's LLM services, focusing on enhancing AI-driven analysis and decision-making capabilities in complex data environments.
- Led the integration and updating of large language models to use OpenAI 1.x.x, GPT-4 Turbo, and GPT-4O, incorporating the latest advancements in AI technologies to boost performance and capabilities.
- Specialized in high-tech sectors, focusing on AI-enhanced software solutions, secure data handling, and infrastructure scalability.

Client: General Motors, Detroit, MI

Aug 2022 – Sep 2023

Role: Sr. Software developer / Python Developer

- Responsible for gathering requirements, system analysis, design, development, testing and deployment.
- Designed and implemented scalable infrastructure as code (IaC) using Terraform, automating the provisioning of resources in cloud environments.
- Proficient in using Terraform for provisioning and managing cloud infrastructure across multiple environments, ensuring consistency and reproducibility.
- Utilized Ansible for dynamic inventory management, enabling real-time adjustments to infrastructure based on current network states.
- Developed web-based open stack applications using Python and Django for large dataset analysis.
- Adept at integrating RESTful APIs and handling asynchronous data operations, ensuring seamless data flow and robust functionality. Proficient in using Vue CLI for efficient project scaffolding, development, and deployment, with a deep understanding of build tools like Webpack and Vite.
- Extensively used regular expressions and core features in Python using lambda, map, reduce etc. and effectively implemented logging feature using python logging library and profiling using cProfile.
- Used Numpy for Numerical analysis for Insurance premium.
- Created an on-premises CI/CD solution using Jenkins and the Pipeline plugin which uses pipeline as code.
- Developed strong knowledge of ML algorithms, metric-driven data set analysis and NLP techniques.
- Agile/SCRUM was used as the project management methodology and JIRA and Confluence were the tools used to keep things in check.
- Clean data and processed third party spending data into maneuverable deliverables within specific format with Excel macros and python libraries such as Numpy, SQLAlchemy and matplotlib.
- Working with backend python automation, CI pipelines, Docker and cloud provisioning /automation.
- Build complex ETL jobs that transform data visually with data flows or by using compute services Azure Databricks, and Azure SQL Database
- Utilized provisioners and post-processors in Packer templates to automate configuration with tools like Ansible, Chef, and Puppet, enhancing image customization and deployment automation.
- Led the migration of legacy applications to PCF, enhancing scalability and performance.
- Automated deployment processes using PCF, achieving a 30% reduction in deployment time.
- Implemented CI/CD pipelines for PCF-hosted applications, ensuring efficient and reliable software delivery.
- Developed & Implemented OAuth2 functionality with Spring Security to secure the REST services/REST API for authentication & authorization and generate & validate the tokens for REST clients to protect the REST services.

- Engineered and maintained multiple serverless solutions using Google Cloud Functions for asynchronous tasks like data processing and real-time file manipulations, leading to a 30% reduction in processing costs.
- Experience in integrating Jenkins with various tools like Maven (Build tool), Git (Repository), Nexus (Artifactory) and implementing CI/CD automation for creating Jenkins pipelines programmatically architecting Jenkins Clusters.
- OAUTH and SSO are used for authentication and Login services.
- Used NLTK and Stanford NLP to process text data and created offline intelligence and used AWS for application deployment and configuration.
- Creating restful web services for Catalog and Pricing with Django MVT, MySQL, and MongoDB.
- Proficient in leveraging the Tkinter Python library to design and implement user-friendly graphical user interfaces for desktop applications.
- Developed tools to automate some base tasks using Shell Scripting, Python.
- Used Jenkins pipelines to drive all micro services builds out to the Docker registry and then deployed to Kubernetes, Created Pods and managed using Kubernetes.
- Responsible for developing ETL pipelines to meet business use cases by using data flows, Azure Data Factory (ADF), Data Lake and Azure Datawarehouse
- Responsible for configuring NoSQL databases like Apache Cassandra and Mongo DB to for increasing compatibility with Django.

Client: CVS Health, Dallas, TX

Dec 2019 – Jul 2022

Role: Full Stack Developer

- Design and Develop Data applications using Python 3.x version and automate deployments using CI/CD procedure via Jenkins and worked closely with Business and involved in gathering requirements.
- Designed and implemented Infrastructure as Code (IaC) using Terraform to provision and manage cloud resources on AWS and Azure.
- Utilized custom scripts and executables in Packer, created multi-platform images, and developed custom builders and provisioners to meet specific deployment requirements.
- Created reusable and modular Terraform templates, facilitating consistent and efficient infrastructure deployment across multiple environments.
- Automated the provisioning of network, compute, and storage resources, reducing manual intervention and configuration errors.
- Involved in migrating the Legacy Application to EPACS application in different stages which includes requirement gathering, design and development. Expertise in writing OOP concepts in Python and involved in using advanced Python modules.
- Designed and developed data management system using MySQL. Involved in Agile Methodologies and SCRUM Process.
- Hands-on experience in using NLTK, Open NLP, and Stanford NLP for Natural Language Processing and Sentimental Analysis.
- Led the implementation of Camunda BPM platform to streamline workflow and business process management across multiple departments, achieving a 30% improvement in process efficiency.
- Created network monitoring and analysis tools using Python (scapy, pyshark) and C# (PacketDotNet, SharpPcap) to capture and analyze network traffic, helping in diagnosing network issues and optimizing performance.
- Setting up Connection Strings and connecting SQL Server AZURE Databases from locally Installed SQL Server Management Studio (SSMS) for Developers
- Collaborated with development and operations teams to integrate Terraform into CI/CD pipelines, streamlining the infrastructure deployment process.
- Expertise in working with different databases like Microsoft SQL Server, Oracle, MySQL, PostgreSQL and Good knowledge in using NoSQL databases MongoDB
- Set-up databases in AWS using RDS, storage using S3 bucket and configuring instance backups to S3 bucket.
- Experience to build CI/CD Pipeline to automate the code release process using Integration tools like Git/SVN, GitHub, Jenkins, Maven and artifact repo.
- Research NLP techniques to construct coherent and clinically comprehensive sentences.

- Loading spilling data using Kafka, Flume and real time Using Spark and Storm.
- Designed and developed scalable Azure APIs using Flask web framework in Python and Integrated with Azure API Management, Logical Apps and other Azure services.
- Involved in creating RESTFUL API's also developed a python level application to consume SOAP API's.
- Created ETL (Extraction, Transformation & Loading) processes in Python performing business rule mappings of relational data into a dimensional model. Involved in pulling data from different databases like Postgres, Oracle, DB2, MySQL and designed a solution to do Balance & Control and Data reconciliation.
- Hands-on technical experience and seasoned Azure Infrastructure and migration to Azure.

Previous Role:

- Myntra – India - Jul 2018 – Dec 2019- Software Engineer

Education

Western Michigan University, Kalamazoo, Michigan

Masters in Computer Science

Jun 2018 – May 2019

Osmania University, Hyderabad India

Bachelors of Engineering in Computer Science

Jul 2013 – Aug 2017